AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2018

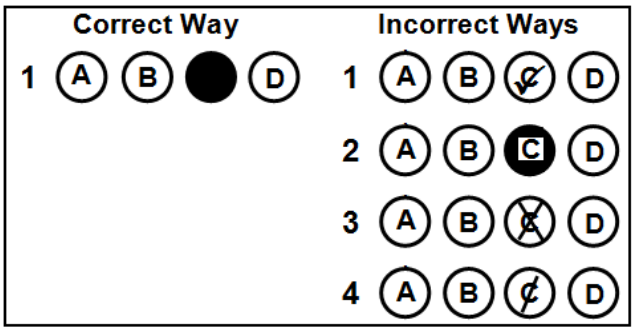
Biology Paper I

Time: 35 minutes Marks: 25

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. A group of researchers worked on the availability of a less expensive fuel for cars. They came up with a conclusion to produce a gas using micro-organisms. To which of the following branches of biology is their research related?

   A. Taxonomy
   B. Entomology
   C. Pharmacology
   D. Biotechnology

2. Which of the following is an example of quantitative observations?

   A. The most complex phase of cell cycle is interphase.
   B. The epithelial cells of trachea are columnar in shape.
   C. The nuclei of leucocytes appear dark under the microscope.
   D. The average life span of thrombocytes is about seven to eight days.

3. A student extracts saliva (pH 7) from the mouth of a mammal, mixes it with some fine pieces of bread and keeps the mixture at 32°C. After three hours, a positive result is obtained, i.e. the pieces of bread disappear from the mixture.

   A positive result can also be obtained by

   A. repeating the procedure at 36°C.
   B. conducting the experiment at pH 2.
   C. using distilled water instead of saliva.
   D. repeating the experiment after boiling saliva.

4. According to the two-kingdom classification system of living organisms, algae were placed in kingdom plantae because of their

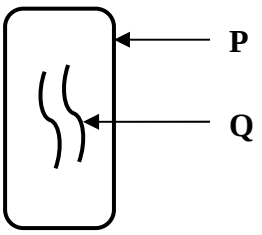
   A. eukaryotic cell.
   B. unicellular body.
   C. mode of nutrition.
   D. mode of reproduction.

5. The similarity between fungi and plants is that both

   A. are prokaryotes.
   B. are heterotrophs.
   C. possess cell wall.
   D. release digestive enzymes.

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6. The given diagram shows the structure of a virus.



Which of the following CORRECTLY identifies **P** and **Q**?

	P	**Q**
A	Lipid bilayer	Nucleic acid
B	Protein coat	Nucleic acid
C	Protein coat	Glycolipid
D	Lipid bilayer	Glycolipid

7. The scientific classification of a plant is as follows.

Order: Asparagales
Genus: Allium
Division: Angiosperm
Family: Amaryllidaceae

Following the rules of binomial nomenclature, the species of the plant would be

A. Allium cepa.
B. Angiosperm cepa.
C. Asparagales cepa.
D. Amaryllidaceae cepa.

8. In plant cells, adenosine triphosphate (ATP) is synthesised by

A. ribosomes.
B. chloroplasts.
C. mitochondria.
D. chromoplasts.

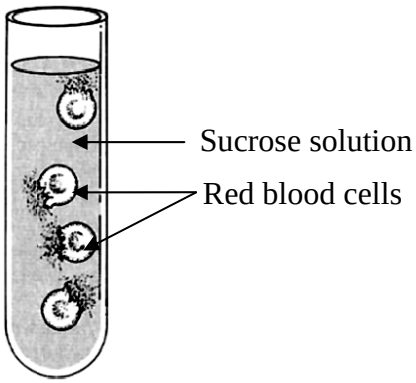
9. The function of smooth endoplasmic reticulum in an animal cell is the

A. storage of enzymes.
B. metabolism of lipids.
C. formation of glucose.
D. synthesis of proteins.

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10. The given diagram shows the condition of human red blood cells when kept in a sucrose solution.

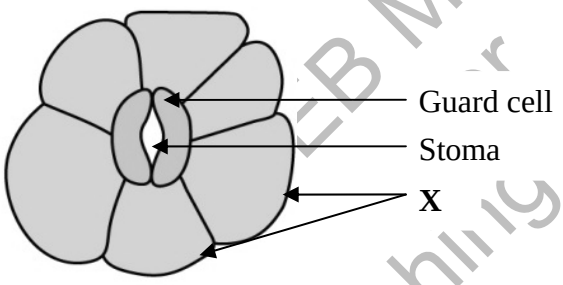
(**Note:** The internal concentration of human red blood cells is 0.9%.)



The concentration of sucrose solution in the given situation would be

A. 3%
B. 5%
C. 0.9%
D. 0.1%

11. The given diagram shows the transverse section of a leaf.



The cells represented by **X** are

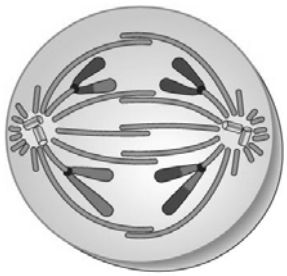
A. phloem cells.
B. epidermal cells.
C. parenchymal cells.
D. collenchymal cells.

12. All of the following statements are TRUE about crossing over in meiosis EXCEPT that it

A. occurs during prophase II.
B. leads to variation in the resulting cell.
C. occurs between two homologous chromosomes.
D. involves exchange of deoxy ribonucleic acid (DNA) fragments.

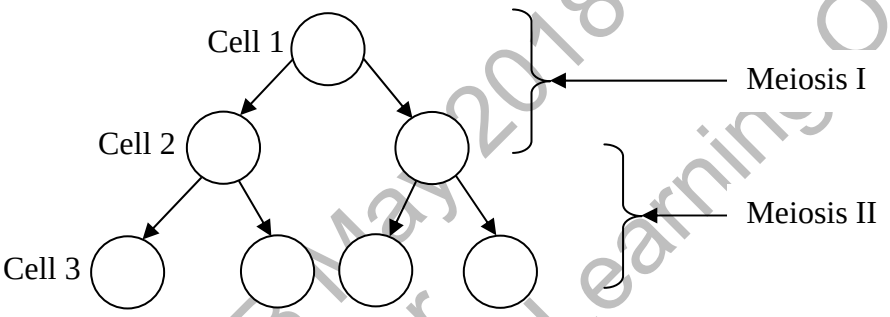
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13. The phase of meiosis in an animal cell illustrated in the given diagram is



A. anaphase of meiosis I.
B. telophase of meiosis I.
C. anaphase of meiosis II.
D. telophase of meiosis II.

14. The given diagram illustrates the process of meiosis.



The haploid cell(s) in the given diagram is/ are

A. cell 1 only.
B. cell 2 only.
C. cell 1 and cell 3.
D. cell 2 and cell 3.

15. Necrosis is a type of cell death characterised by

A. formation of cell blebs.
B. condensation of chromatin.
C. leakage of lysosomal enzymes.
D. breaking of nuclear membrane.

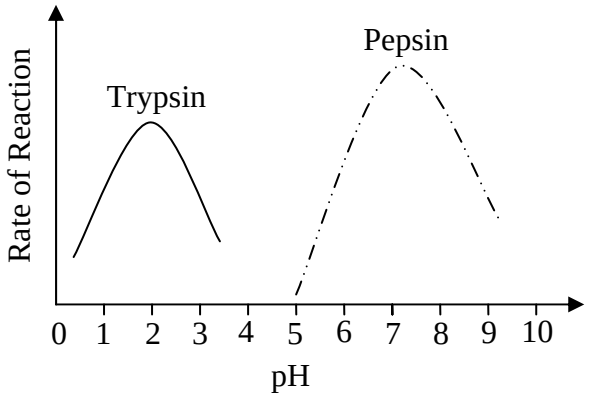
16. Asif goes for a morning walk where the temperature of the surrounding is 13°C.

In this situation, enzymes in Asif's body would work best at a body temperature of

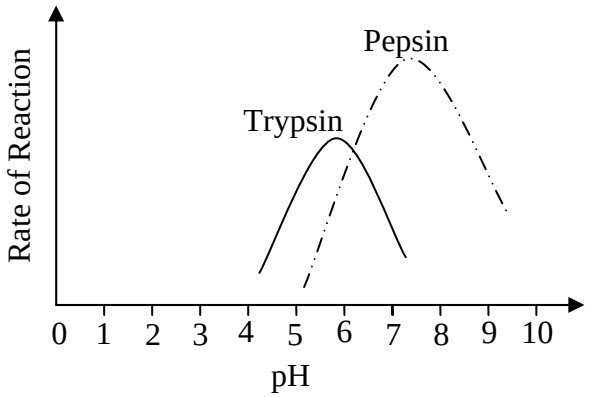
A. 13°C
B. 23°C
C. 30°C
D. 37°C

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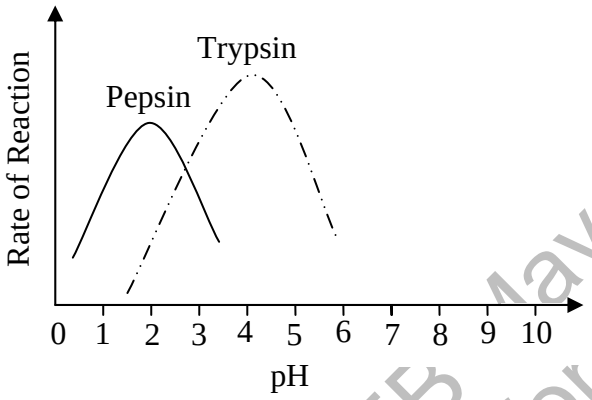
17. Which of the given graphs CORRECTLY represents the activity of pepsin and trypsin at their respective pH?



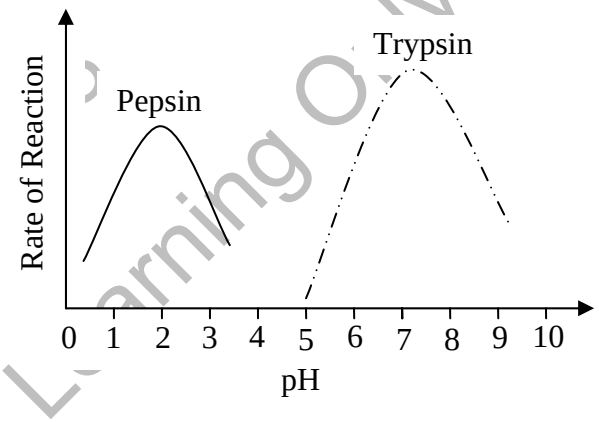
A



B

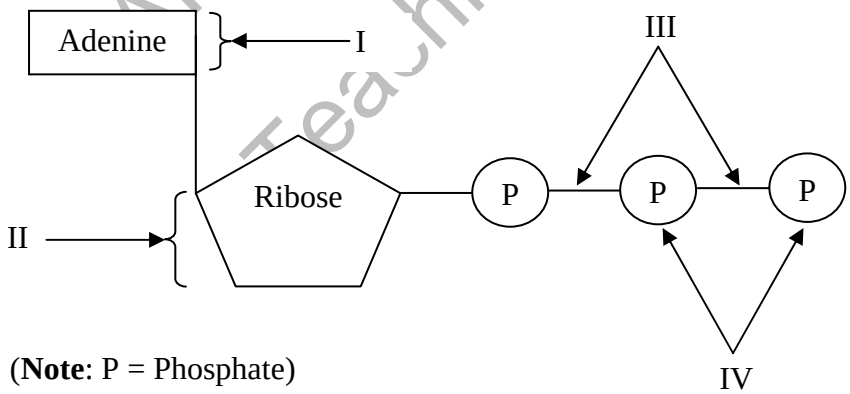


C



D

18. The given diagram illustrates the chemical structure of adenosine tri phosphate (ATP).



(**Note**: P = Phosphate)

Which labelled part of ATP provides energy when broken?

A. I
B. II
C. III
D. IV

19. The reduction of nicotinamide adenine dinucleotide phosphate ($NADP^+$) takes place during the light reactions of photosynthesis by

A. gaining two electrons and one hydrogen ion.
B. gaining one electron and two hydrogen ions.
C. removing two electrons and one hydrogen ion.
D. removing one electron and two hydrogen ions.

20. The similarity between aerobic and anaerobic respiration is the

A. complete oxidation of glucose.
B. production of equal amount of energy.
C. involvement of mitochondria in reactions.
D. occurrence of oxidation reduction reactions.

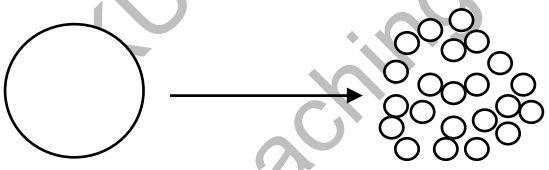
21. The given table shows the daily diet of Sania.

Breakfast	Lunch	Dinner
A fried egg and 2 slices of bread	Boiled rice, chicken and fruit salad	Pasta and vegetables

This diet could be considered balanced because it includes

A. less fats.
B. three meals.
C. variety of food.
D. more carbohydrates.

22. The given diagram illustrates a process occurring in the digestive system of human beings.



Fat drop Fat droplets

This process is called

A. assimilation.
B. deamination.
C. emulsification.
D. chemical digestion.

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23. A patient has the following symptoms.

- Pain in abdominal region
- Difficulty in expelling faeces
- Infrequent bowel movements

The patient is MOST likely to be suffering from

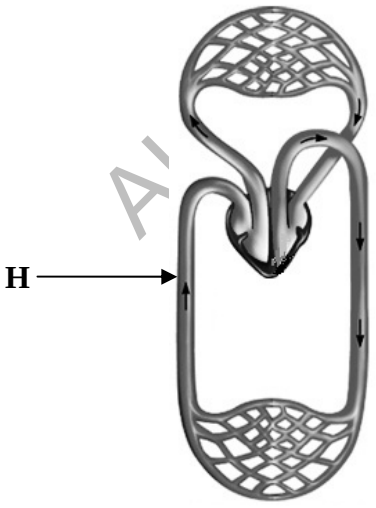
A. ulcer.
B. typhoid.
C. diarrhoea.
D. constipation.

24. Saima, who has blood group **B**, needs blood.

People with which of the following blood groups can donate blood to Saima?

	Blood group A	**Blood group B**	**Blood group AB**	**Blood group O**
A	Yes	No	Yes	No
B	Yes	No	No	Yes
C	No	Yes	Yes	No
D	No	Yes	No	Yes

25. The given diagram shows double circulation of blood in a mammal.



The blood vessel **H** represents

A. vena cava.
B. dorsal aorta.
C. femoral artery.
D. pulmonary vein.

END OF PAPER

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